Brokers and cartridges

The broker acts like a single point of contact for all things related to the application. The command-line tools and Web console both connect to this broker with a REST-based API. Cartridges, on the other hand, are again containers that run the architectural and framework components of an application and provide enhanced features for the application. The cartridges can run on one or more gears, and different cartridges for components like PHP, JBoss, Ruby and MySQL are available. The two types of cartridges available are framework and embedded cartridges. The embedded cartridges are those that perform support functions like monitoring, database management, performance metrics etc. If you are well acquainted with this platform, you should be able to write your own cartridges to add custom features to your application.

Districts

Districts are a set of nodes, which act like resource pools that allow gears to be moved between the nodes within the district, so as to allow for load balancing to avoid overloading of a particular node. Districts are optional, but generally recommended for all applications that need to work in production environments.

Developers should also be aware of other application components like namespace, application name, aliases, the Git repository, etc. OpenShift also provides for manual and automatic scaling of Web applications, both of which allow you to add multiple instances to tackle the increase in demand. While automatic scaling is easy and convenient, it takes an extra cartridge called HAProxy or High Availability Proxy, which monitors all incoming traffic and assigns an extra gear whenever there is a spurt in demand for the Web application.

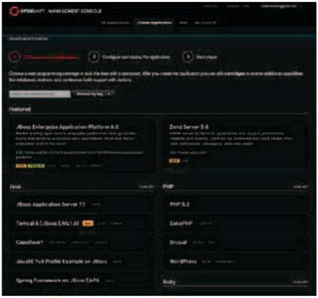

Figure 1: Choosing the type of application for creation

Supported languages and frameworks

The supported languages and environments for OpenShift include Node.js, Ruby, Python, PHP, Perl, and surprisingly, Java. There are very few cloud application platforms that are associated with such a large variety of environments, and particularly with Java, which is usually restricted to enterprise applications. It is actually the first such platform ever to support Java EE in its entirety. Among the other frameworks it supports are Ruby on Rails, Django, CakePHP, CodeIgniter, etc. Among the database options are MongoDB, PostgreSQL and MySQL, which are currently available with phpMyAdmin and rockMongo as the consoles, which can also be accessed via SSH.

Installing the client

The client tools can be installed on all OSs including Windows, Mac OS X and Linux. Documentation is available for distributions like Ubuntu, Debian, Fedora, OpenSUSE and RHEL 6. The installation procedure is, however, similar across all platforms. Before proceeding to install, the required dependencies are Ruby, Ruby Gems and Git. On Ubuntu, for example, these can be satisfied with a *sudo apt-get install ruby-full rubygems git-core* command. Finally, the client can be installed with *sudo gem install rhc*.

Now, the software has been installed by default in the */var/lib/gems/1.8/bin* directory, which can be made universally accessible by adding it to the *PATH* variable and creating a symbolic link to it in the */usr/bin* directory:

```
sudo export PATH=/var/lib/gems/1.8/bin:$PATH
sudo ln -s /var/lib/gems/1.8/bin/rhc* /usr/bin/
```

To start configuring authentication, you can either generate SSH keys manually or preferably use the set-up wizard, which you can invoke with *rhc setup;* it will guide you through creating the SSH keys and will later ask you to upload the generated public key to the OpenShift server from the console.

Deploying applications from the control panel

Working on the Web console is pretty easy, and is just a matter of registering and working with graphical wizards that guide you through every step. It starts with registering your account and defining the namespace of the application. This is immediately followed by the *Create Application* wizard, which is a three-step process of choosing the application type, entering the configurations and confirming the creation of the application (Figures 1 and 2). OpenShift even provides for preconfigured applications like WordPress and Drupal, which require minimal effort on the part of the user. After creating the application, the options related to